Digitalizing Midwifery Care: VR & Mobile Application Solutions for Maternal Education

Project Proposal Report

Kumarage M.B.S.I.E – IT22606556

BSc (Hons) in Information Technology – Interactive Media

Department of Information Technology

Sri Lankan Institute of Information Technology

June 2025

Machine Learning–Based Interactive Mobile App For Early Risk Detection and Midwifery Support

Project Proposal Report

Kumarage M.B.S.I.E – IT22606556

BSc (Hons) in Information Technology – Interactive Media

Department of Information Technology

Sri Lanka Institute of Information Technology

June 2025

Declaration, copyright statement and the statement of the supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Kumarage M.B.S.I. E	IT22606556	

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Mr. Didula Thanaweera Arachchi

Signature of the supervisor: Date: 8/22/2025

Abstract

This research proposes the design and development of an interactive mobile application that supports Sri Lankan midwives in antenatal and intrapartum care through real-time machine learning–based maternal risk prediction, interactive data visualization, and multimodal alerts. The system integrates maternal vital data such as blood pressure, blood sugar, fetal heart rate, and hemoglobin levels to predict early risk conditions including anemia and pregnancy-related complications.

Unlike existing maternal health applications that are either mother-focused or guideline-only tools, the proposed system is specifically designed for midwives and midwifery students. It provides real-time dashboards, explainable machine learning outputs, and voice and haptic feedback to improve decision-making speed, accuracy, and confidence. The application is culturally adapted with Sinhala, Tamil, and English language support and is optimized for low- and mid-range mobile devices commonly used in Sri Lanka. By combining mobile development, interactive media, and applied machine learning, this project aims to enhance maternal safety and strengthen midwifery competency.

Table of Contents

Abstract ii

Table of Contents iii

List of Abbreviations v

Introduction 1

- Background & Literature survey 1
- Research Gap 2
- Research Problem 4

Objectives 4

- Main Objectives 4
- Specific Objectives 4

Methodology 5

- Functional Requirements 6
- Non-Functional Requirements 6
- System Requirements Hardware Requirement 7

Description of Personal and Facilities 7

- Supervisors 7
- External Supervisor 7
- Facilities 7
- Sample of Midwifery Students 7
- Device Access 7

Budget and Budget Justification 8

Reference List 8

Table of Figures

Figure 1 Midwifery Care Phases (Antenatal, Intrapartum) 2
Figure 2 Antenatal Monitoring Parameters (BP, FHR etc.) 2
Figure 3 Individual System Diagram 5

List of Tables

Table 1 Research Gap 4
Table 2 System Gap 4
Table 3 Technologies Used 6
Table 4 Budget Table 8

List of Abbreviations

Abbreviation	Description
mHealth	Mobile Health
BP	Blood Pressure
FHR	Fetal Heart Rate
ANC	Antenatal Care
PPH	Postpartum Hemorrhage
API	Application Programming Interface
UI	User Interface
UX	User Experience
SQL	Structured Query Language
DB	Database
ML	Machine Learning
WHO	World Health Organization
FHB	Family Health Bureau (Sri Lanka)
MOH	Ministry of Health (Sri Lanka)

Introduction

Midwives play a critical role in Sri Lanka's maternal healthcare system, particularly in antenatal monitoring and early detection of pregnancy-related risks. Although Sri Lanka has achieved significant reductions in maternal mortality, preventable complications such as anemia, hypertension, and delayed emergency response remain challenges, especially in rural settings.

With the increasing availability of digital health data, there is an opportunity to enhance midwives' decision-making using intelligent mobile tools. However, most existing research on maternal risk prediction using machine learning remains confined to academic models without real-time mobile deployment or integration into midwifery workflows. This project responds to that gap by proposing a machine learning–based interactive mobile application that delivers real-time risk predictions, intuitive visualizations, and actionable alerts directly to midwives.

Background & Literature survey

Recent studies demonstrate the effectiveness of machine learning models such as Random Forest, SVM, XGBoost, and deep learning architectures in predicting maternal health risks with accuracies exceeding 85%. Research by Assaduzzaman et al. (2023) and Harnal et al. (2025) highlights improved predictive performance using advanced validation techniques, while Uma Maheswari et al. (2024) emphasizes handling data imbalance and explainability using SHAP.

Despite these advancements, existing solutions lack deployment in real-time mobile applications and do not address midwives' usability needs. Furthermore, explainability is often limited to offline analysis, and no prior systems incorporate voice or haptic feedback to support rapid decision-making. The proposed system bridges this gap by embedding lightweight ML models within a mobile app, supported by interactive dashboards and multimodal alerts.

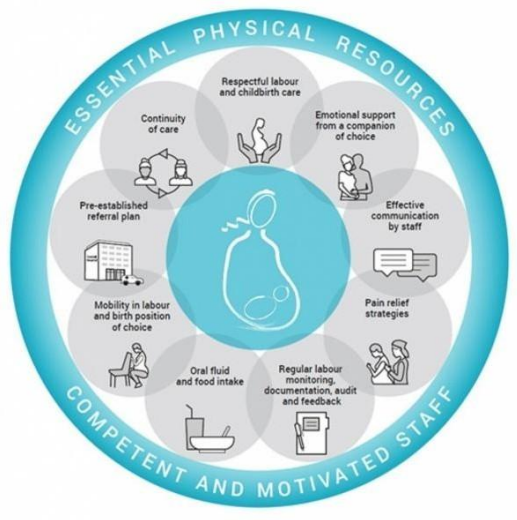

Figure 1 Midwifery Care Phases (Antenatal, Intrapartum)

Figure 2 Antenatal Monitoring Parameters (BP, FHR etc.)

Research Gap

Although machine learning techniques for maternal risk prediction are well researched, the following gaps remain:

- Lack of midwife-focused mobile applications integrating ML predictions
- Absence of real-time interactive data visualization
- No use of voice or haptic feedback for urgent alerts
- Limited cultural and language adaptation for Sri Lankan users
- Minimal usability and clinical simulation testing with real users

The proposed system uniquely addresses all these gaps.

Table 1 Research Gap

Research Paper	Research Paper Discussion	Research Gap
Assaduzzaman et al. (2023)	Built ML classifiers (SVM, Random Forest) achieving ~89% accuracy for early maternal risk prediction.	Not integrated into a midwife-focused mobile app; no real-time interface.
Harnal et al. (2025)	Compared percentage split vs. k-fold cross-validation; k-fold improved accuracy by ~5%.	Lacked interactive explainability or deployment in clinical settings.
Uma Maheswari et al. (2024)	Applied SMOTE and SHAP with XGBoost to handle imbalance and explain feature importance; recall improved by 12%.	No live dashboards or clinical workflow integration; explainability only post-hoc.
Rahman & Alam (2023)	Used CNN/LSTM models with LIME for deep learning risk prediction; achieved 91% F1-score.	High computational load; no real-time on-device inference or voice/haptic alerts.
Ravi et al. (2022)	Analyzed time-series trajectories of maternal risk, identifying distinct risk patterns over pregnancy.	Did not embed predictive models into a user interface; no mobile deployment or midwife-centered design.

Table 2 System Gap

Feature	Proposed System	Assaduzzaman et al. (2023)	Harnal et al. (2025)	Uma Maheswari et al. (2024)	Rahman & Alam (2023)	Ravi et al. (2022)
Uses machine learning	✓	✓	✓	✓	✓	✓
Includes data visualization	✓	✗	✗	✗	✗	✗
Focus on midwives	✓	✗	✗	✗	✗	✗
Includes real-time risk detection	✓	✓	✓	✓	✓	✓
Voice & Haptic feedback	✓	✗	✗	✗	✗	✗
Supports local cultural adaptation	✓	✗	✗	✗	✗	✗
Prototype tested with users	✓	✓	✗	✗	✗	✗

Research Problem

Midwives in Sri Lanka lack access to intelligent, real-time digital tools that transform maternal health data into actionable insights during antenatal care.

Problem Statement:

How effective is an interactive mobile application with machine learning and data visualization in improving midwives' decision-making in maternal care?

Objectives

Main Objectives

To develop and evaluate a machine learning–based interactive mobile application that improves antenatal risk detection and decision-making among midwives.

Specific Objectives

1. Design and integrate lightweight supervised machine learning models for real-time maternal risk prediction.
2. Develop interactive dashboards to visualize maternal health trends and risk levels.
3. Implement voice and haptic alerts to support rapid clinical response
4. Ensure cultural relevance through multilingual (Sinhala, Tamil, English) UI design.\

5. Evaluate system usability, predictive accuracy, and decision impact through midwife testing and clinical simulations

Mobile Application Development

• Cross-platform (Flutter / React Native).
• Interactive UI/UX for maternal monitoring & birth-phase modules.
• Local language support and offline accessibility.

Data Analysis & Dashboard Design

• Maternal health data stored in database (SQL/Firebase).
• Visualization of trends (BP over time, fetal growth).
• Alerts for abnormal values.
• Student training performance reports.

Methodology

The project follows a mixed-method development and evaluation approach:

1. Requirement Gathering – midwife interviews and review of FHB guidelines.
2. Data Collection & Preprocessing – anonymized maternal datasets.
3. ML Model Development – training and evaluating supervised models; conversion to TensorFlow Lite.
4. Mobile App Development – Flutter/React Native implementation with dashboards and alerts.
5. Explainability & Visualization – SHAP/LIME integration and interactive charts.
6. Testing & Validation – usability testing (SUS), clinical simulation trials, and performance evaluation.

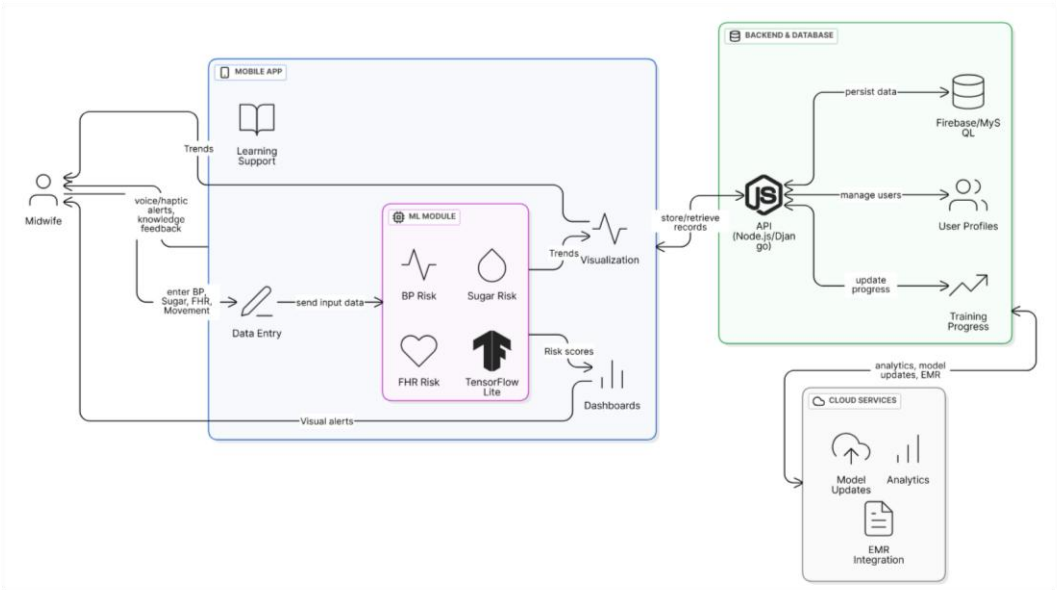

Figure 4 Individual System Diagram

Table 3 Technologies Used

Technology	Description
Flutter / React Native	Cross-platform mobile development framework used to build the mobile application (Android & iOS).
Firebase / MySQL	Backend database solutions to store maternal health data, training data, and user profiles.
RESTful API (Node.js / Django)	Middleware for communication between mobile app and backend database.
Python (Pandas, Scikit-learn)	Used for data analysis, trend prediction, and generating insights from maternal health data.
Chart.js / Power BI	Data visualization tools for creating dashboards and performance analytics.
Figma	UI/UX design tool used for prototyping the mobile application.
Android Studio / VS Code	Development environments for coding, debugging, and testing.
GitHub	Version control system for managing collaborative development.

Functional Requirements

1. Input and tracking of maternal health data (BP, FHR, blood sugar, Hb).
2. Real-time ML-based risk prediction.
3. Interactive dashboards with trend visualization.
4. Voice and haptic alerts for abnormal conditions.
5. Multi-language and offline support.

Non-Functional Requirements

1. Secure handling of sensitive health data.
2. Fast on-device ML inference.
3. High usability and accessibility.
4. Scalability for multiple clinics and users.

System Requirements

Hardware Requirement

1. Smartphones (Android/iOS).
2. Mid-range laptops for development.

Software Requirements

1. Flutter/React Native framework.
2. Firebase / SQL database.
3. Power BI / Chart.js for dashboards.

Table 4 Work Breakdown Chart

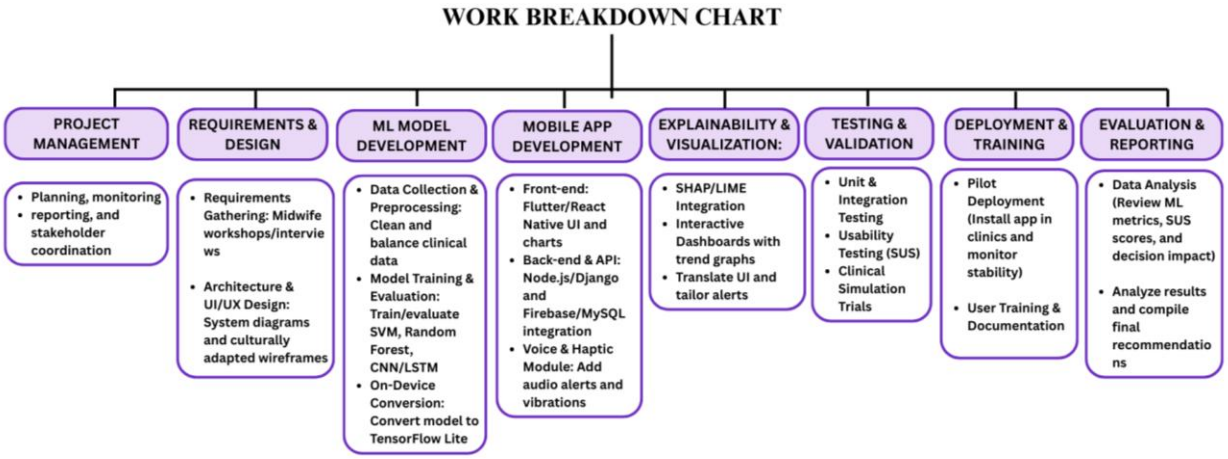

Table 5 Grantt Chart

Phase	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Literature Review & Benchmarking							
AI Model Development (CVA, CI, Posture)							
Data Preprocessing & Testing (OpenCV)							
UI/UX Design (Figma, Audio Guidance)							
App Development (Flutter, SQLite)							
Offline Functionality & Localization							
Midwife Usability Testing & Feedback							
Final Validation & Play Store Prep							

Description of Personal and Facilities

Supervisors
Our supervisor Mr. Didula Thanaweera Araghchi and Co-Supervisor Mrs. Narmada Gamage, who are experts in the technical field, will provide guidance in software development and technology integration.

External Supervisor
Our external supervisor is a midwife who has been in the field for over 26 years. She shared real experiences, how she manages mothers before, during and after birth. Her knowledge helped us understand the real problems midwives face daily. With her guidance, we were able to plan the app feature more practically and usefully.

Facilities
The project is carried out with support from the Humanities and Nursing Departments of SLIIT. Access is provided to medical books, maternal health guidelines, and computer labs for app development and testing. This setup ensures that the mobile app is developed in line with real-world midwifery practices in Sri Lanka.

Sample of Midwifery Students
A group of 15–20 midwifery and nursing students will take part in the testing phase. They will use the app, give feedback on its features, and help improve its design and content. This hands-on testing also gives students practical learning opportunities.

Device Access
The app will be tested on commonly used Android smartphones, including low and mid- range devices. This ensures it works well for public health midwives in both urban and rural areas. The testing will also cover local language support, offline access, and data entry functions.

Budget and Budget Justification

Table 6 Budget Table

Expenses	
Requirement	Cost (Rs.)
Mobile Devices for Testing (Android + iOS)	180,000
Backend Server Hosting (1 year)	60,000
Firebase / Cloud Database (annual)	45,000
Software Licenses (Figma, IDEs, API tools)	30,000
Data Analysis Tools (Python Libraries, Power BI Pro)	50,000

Reference List

[1] Assaduzzaman, M., Al Mamun, A., & Hasan, M. Z. (2023, January). Early prediction of maternal health risk factors using machine learning techniques. In 2023 International Conference for Advancement in Technology (ICONAT) (pp. 1-6). IEEE. Link

[2] Harnal, S., Sharma, A., Malhotra, J., Sharma, S., & Grover, M. (2025, January). Predicting Maternal Health Risk Using Machine Learning Models and Comparing The Performance Of Percentage Split And K-Fold Cross Validation. In 2025 1st International Conference on AIML-Applications for Engineering & Technology (ICAET) (pp. 1-8). IEEE. Link

[3] Uma Maheswari, B., Dixit, A., & Karn, A. K. (2024). Machine Learning Algorithm for Maternal Health Risk Classification with SMOTE and Explainable AI. In 2024 IEEE 9th International Conference for Convergence in Technology (I2CT) (pp. 1-6). IEEE. Link

[4] Rahman, A., & Alam, M. G. R. (2023). Explainable AI based Maternal Health Risk Prediction using Machine Learning and Deep Learning. In 2023 IEEE World AI IoT Congress (AIIoT). IEEE. Link

[5] Ravi, A., Joshi, S. P., Kodipalli, A., & Kamal, S. (2022). Analysis of maternal health risk using computational models. In 2022 International Conference on Smart and Sustainable Technologies in Energy and Power Sectors (SSTEPS). IEEE. Link

[6] Al-Bahadly, I., & Ali, K. (2024). Ensemble machine learning framework for predicting maternal health risk classification. Scientific Reports, 14, Article 71934. https://www.nature.com/articles/s41598-024-71934-x

[7] Debnath, S., & Das, A. (2025). Automated Pregnancy Risk Level Prediction Using Advanced Machine Learning and Deep Learning Algorithm. International Journal of Computer Applications, 186(77), 24–29. https://ijcaonline.org/archives/volume186/number77/automated-pregnancy-risk-level-prediction-using-advanced-machine-learning-and-deep-learning-algorithm/

[8] Mukwende, M., et al. (2025). Machine Learning Models to Predict Risk of Maternal Morbidity and Mortality Outcomes: Scoping Review. Journal of Medical Internet Research, 27(1), e68225. https://www.jmir.org/2025/1/e68225

[9] Gyamfi, V., et al. (2025). The mHealth clinical decision-making tools for maternal and perinatal health: A systematic review. PLoS Medicine, 22(4), e11042025. https://pmc.ncbi.nlm.nih.gov/articles/PMC11042025/

[10] Bodnar, L. M., et al. (2024). Public Maternal Health Dashboards in the United States: Descriptive Evaluation. Journal of Medical Internet Research, 26, e56804. https://www.jmir.org/2024/1/e56804/